Samy Lahlou Kamal

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samylahlou.com

Education

August 2022 – present **McGill University** **Montréal, Canada**
Bachelor of Science - Honours Mathematics.
GPA: 3.81/4.0

September 2015 – June 2022 **Lycée Lyautey** **Casablanca, Morocco**
French Baccalaureate with High Honors (Mention Très Bien).
Specialities: Mathematics & Computer Science
Options: Expert Mathematics & Music

Research Experience

May 2025 – present **Research Project on Special Values of L-functions**
Supervisor: Professor Henri Darmon (McGill University)

- Currently working on the history of L-functions (reading the original books and papers), and their special values.
- Presented my work in the USRA Number Theory seminar.
- Started writing an expository paper on the detailed history of L-functions and the mathematicians that contributed to that subject (L. Euler, P.L. Dirichlet, B. Riemann, R. Dedekind, E. Hecke, ...): [pdf (unfinished)]
- Funded by the NSERC (Natural Sciences and Engineering Research Council of Canada).

May 2024 – August 2024 **Research Project on the Ergodic Theorems**
Supervisor: Professor Anush Tserunyan (McGill University)

- Studied Measured Group Theory with applications to the Ergodic Theorems.
- Wrote an expository paper on the ergodic theorems with a focus on Pr. Tserunyan's new proof of the Birkhoff Ergodic Theorem: [pdf]
- This project was part of the Honours Research Project class MATH 470.

January 2024 – April 2024 **Directed Reading Program on p-adic Numbers**
Supervisor: Doctoral student Hazem Hassan (McGill University)

- Studied p-adic numbers and p-adic Zeta-Functions by reading the book *p-adic Numbers, p-adic Analysis, and Zeta-Functions* by Neal Koblitz.
- Presented my learnings in front the other Directed Reading Program students.
- Wrote a report presenting p-adic numbers, their construction and p-adic interpolation.

Professional Experience

September 2025 – present **Teaching Assistant for the course History & Philosophy of Mathematics**
McGill University, Montréal, QC
Instructor: Professor Henri Darmon, Course Code: MATH 338, 135 students

- Conducted tutorial sessions (three hours a week) and exam review sessions to go over the course content, additional content, and answer students' questions.
- Wrote a suitable set of notes based on the students' questions to complete the ressources available to the students.
- Planned one-on-one meetings with students who wanted to.

January 2024 – April 2024 **Grader for the course Discrete Structures**
McGill University, Montréal, QC
Instructor: Mr. Jeremy Macdonald, Course Code: MATH 240, 365 students

- Graded assignments (one every two weeks) for the course Discrete Structures.

May 2023 – September 2024 **Tutor in Mathematics**
McGill University, Montréal, QC

- Planning tutoring sessions (private or in group) for 1st year students in Mathematics.

Projects

January 2025 **Expository Paper on the Divergence of the Harmonic Series**

- Wrote a self-contained paper presenting the importance of rigor in mathematics through the detailed study of the divergence of the Harmonic Series: [pdf]

December 2024 **Published Historical Article on Fourier Analysis**

- Researched on the origins and impacts of Fourier analysis on modern mathematics through the original works of the great mathematicians: J. Fourier, P.L. Dirichlet, B. Riemann, G. Cantor.
- Co-authored (along with Nisrine Sqalli) a historical paper presenting our findings: [pdf]
- Published a condensed version in the Delta Epsilon journal: [pdf]

December 2024 – present **Personal Website**

- Coded a personal website by learning and using HTML and CSS: [samylahlou.com]
- Post regularly my personal solutions to textbook exercises, articles that I write and reviews of textbooks and books related to mathematics.

Awards

NSERC Undergraduate Student Research Award – $8950

Coding Skills

Programing Languages

- Python
- Java
- C
- Bash
- Assembly
- SQL
- MATLAB
- SageMath

Markup Languages

- LaTeX
- HTML
- CSS

Languages

English (Fluent) French (Native) Arabic (Native)